

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1197

Mattoon East 138 kV
December 31, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1197. The Interconnection Customer proposes a 217.3 MW solar photovoltaic facility interconnecting at the Mattoon East 138 kV substation in the West Virginia area. The request was received on April 30, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 360 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$99,250,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Goat Rock-Panda 138 KV line
Contingency: Panda-Northern Steel Casting 138 KV line
Loading: 187.8% (Rating: 188 MVA)
DFAX: 16.0%
Pre-project loading: 169.3%
2. Facility: Northern Steel Casting-Goat Rock 138 KV L
Contingency: Panda Substation-Northern Steel Casting 138 KV
Loading: 180.2% (Rating: 199 MVA)
DFAX: 16.0%
Pre-project loading: 162.7%
3. Facility: Panda-Northern Steel Casting 138 KV
Contingency: Goat Rock-Northern Steel Casting 138 KV
Loading: 175.4% (Rating: 226 MVA)
DFAX: 17.7%
Pre-project loading: 158.3%
4. Facility: Carlsbad Watfield-Menlo Substation 138 KV L
Contingency: Menlo Substation-W Woodland 138 KV line
Loading: 168.2% (Rating: 205 MVA)
DFAX: 15.8%
Pre-project loading: 151.5%
5. Facility: E. Ripley-Carlsbad Waterfield 138 KV
Contingency: Menlo-Wet Woodland 138 KV
Loading: 154.5% (Rating: 219 MVA)
DFAX: 15.8%
Pre-project loading: 138.8%

6. Facility: E Ripley-West Fork 138 KV L
Contingency: Menlo-West Woodland 138 KV
Loading: 145.2% (Rating: 227 MVA)
DFAX: 15.8%
Pre-project loading: 130.1%
7. Facility: Rockaway Beach Substation-Hampshire Paper 138 KV L
Contingency: Iron Grove-Waterton 138 KV
Loading: 121.8% (Rating: 303 MVA)
DFAX: 27.6%
Pre-project loading: 102.0%
8. Facility: Mattoon East-Rockaway Beach 138 KV line
Contingency: Waterton-Iron Grove 138 KV
Loading: 121.4% (Rating: 307 MVA)
DFAX: 27.6%
Pre-project loading: 101.8%
9. Facility: Panda-Menlo 138 KV L
Contingency: Fishkillplains-Dillard-Orange 345 KV
Loading: 116.5% (Rating: 383 MVA)
DFAX: 18.4%
Pre-project loading: 106.0%
10. Facility: Euleess-Tank Hill Substation 138 KV
Contingency: Carlsbad Waterfield-Menlo 138 KV Line
Loading: 115.9% (Rating: 230 MVA)
DFAX: 13.0%
Pre-project loading: 103.5%
11. Facility: Euleess-W Woodland 138 KV
Contingency: Carlsbad Waterfield-Menlo 138 KV
Loading: 115.8% (Rating: 233 MVA)
DFAX: 13.0%
Pre-project loading: 103.7%
12. Facility: Solana-Northern Steel Casting 138 KV Line
Contingency: Dillard-Orange-Fishkill Plains 345 KV
Loading: 115.4% (Rating: 397 MVA)
DFAX: 18.4%
Pre-project loading: 105.4%
13. Facility: Menlo-W Woodland 138 KV Line
Contingency: Carlsbad Waterfield-Menlo 138 KV
Loading: 115.3% (Rating: 237 MVA)
DFAX: 13.0%
Pre-project loading: 103.3%
14. Facility: Solana Substation-Hampshire Paper 138 KV line
Contingency: Fishkill Plains-Dillard-Orange 345 KV
Loading: 115.1% (Rating: 403 MVA)
DFAX: 18.4%
Pre-project loading: 105.2%
15. Facility: Union Vly-Iron Grove 138 KV
Contingency: Rockaway Beach-Mattoon East 138 KV
Loading: 105.3% (Rating: 204 MVA)
DFAX: 8.4%
Pre-project loading: 96.4%
16. Facility: Potomac Yard-Union Vly 138 KV
Contingency: Mattoon E-Rockaway Bach 138 KV

Loading: 102.5% (Rating: 251 MVA)
 DFAX: 14.5%
 Pre-project loading: 90.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Goat Rock-Panda 138 KV line	Rebuild line with higher-capacity conductor	\$9,826,000	89% (\$8,731,000)
Northern Steel Casting-Goat Rock 138 KV L	Rebuild line with higher-capacity conductor	\$8,990,000	89% (\$7,988,000)
Northern Steel Casting-Panda 138 KV	Rebuild line with higher-capacity conductor	\$3,319,000	89% (\$2,949,000)
Carlsbad Watrfield-Menlo Substation 138 KV L	Rebuild line with higher-capacity conductor	\$3,946,000	91% (\$3,591,000)
E. Ripley-Carlsbad Waterfield 138 KV	Rebuild line with higher-capacity conductor	\$5,958,000	91% (\$5,421,000)
E. Ripley-W Fork 138 KV	Rebuild line with higher-capacity conductor	\$6,276,000	91% (\$5,711,000)
Hamshire Paper-Rockaway Beach 138 KV	Rebuild line with higher-capacity conductor	\$15,972,000	94% (\$15,040,000)
Mattoon East-Rockaway Beach 138 KV line	Rebuild line with higher-capacity conductor	\$10,680,000	94% (\$10,057,000)
Panda-Menlo 138 KV L	Rebuild line with higher-capacity conductor	\$6,604,000	89% (\$5,886,000)
Tank Hill-Euleess Substation 138 KV	Rebuild line with higher-capacity conductor	\$4,219,000	86% (\$3,609,000)
Euleess-W Woodland 138 KV	Rebuild line with higher-capacity conductor	\$8,889,000	86% (\$7,605,000)
Solana-Northern Steel Casting 138 KV Line	Rebuild line with higher-capacity conductor	\$4,895,000	89% (\$4,362,000)
Menlo Substation-W Woodland 138 KV Line	Rebuild line with higher-capacity conductor	\$7,944,000	86% (\$6,796,000)
Solana-Hampshire Paper 138 KV	Rebuild line with higher-capacity conductor	\$6,218,000	89% (\$5,542,000)
Union Vly-Iron Grove 138 KV	Reconductor line	\$2,762,000	100% (\$2,762,000)
Potomac Yard-Union Vly 138 KV	Reconductor line	\$3,200,000	100% (\$3,200,000)
	Total allocated to project		\$99,250,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.